Patient Churn Prediction

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TL;DR

Random Forest classification was used to predict patient churn given a 500-observation synthetic data set. Patients' age, satisfaction with care, and healthcare total bill amount are key predictors of patient churn. Churn initially decreases with age and satisfaction and initially increases with bill amount.

Data and Methods

Data Overview

- 500 synthetically generated observations from Kaggle for hospital churn prediction
- “Churn” defined as whether patient has ceased using the hospital's services
- Each row is a single individual with information on age, visits missed, insurance type, bill amount, and more

	Patient_ID	Age	Gender	Tenure_Months	Visits_Last_Year	Chronic_Disease	Insurance_Type	Satisfaction_Score	Total_Bill_Amount	Missed_Appointments
0	1	56	Female	2	3	No	Government	2.5	12252.96	9
1	2	69	Male	10	3	Yes	Government	2.6	25862.01	4
2	3	46	Female	56	10	No	NaN	2.8	5659.13	4
3	4	32	Male	30	4	Yes	Government	4.1	19533.31	5
4	5	60	Male	50	19	No	Private	4.6	24639.52	5

Methods

1. Data cleaning and exploratory data analysis in Python
2. Creation of 80/20 train-test split to limit data leakage during modeling
3. Model pipeline creation, including preprocessing, scaling, transformation
4. 5-fold cross-validation grid search for hyperparameter tuning
5. Training of final model and application to test set to evaluate model performance

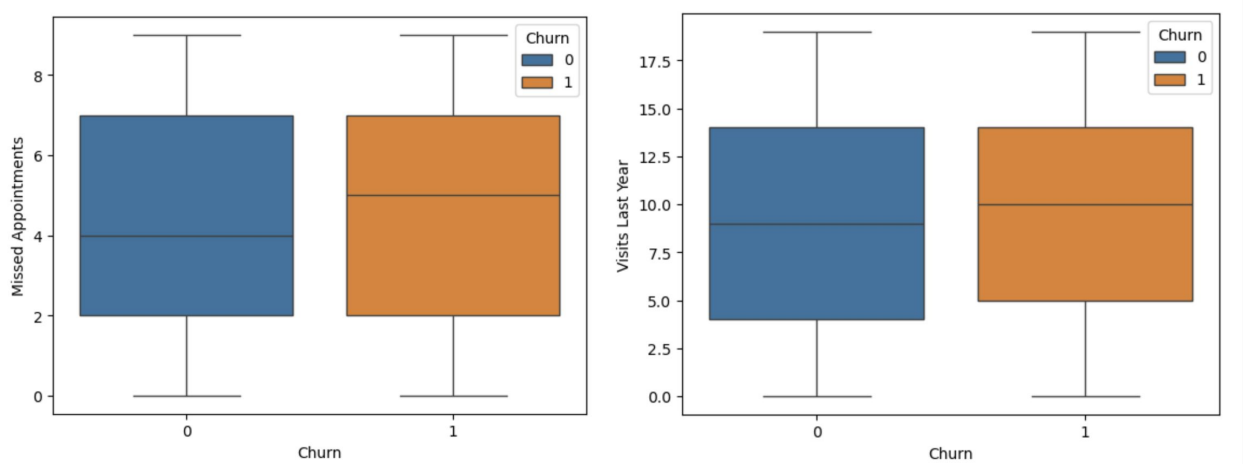
Data Cleaning

Insurance_Type
Government
Government
Government
Private
NaN

NaN values in Insurance_Type column were replaced with "None" to indicate patients were uninsured

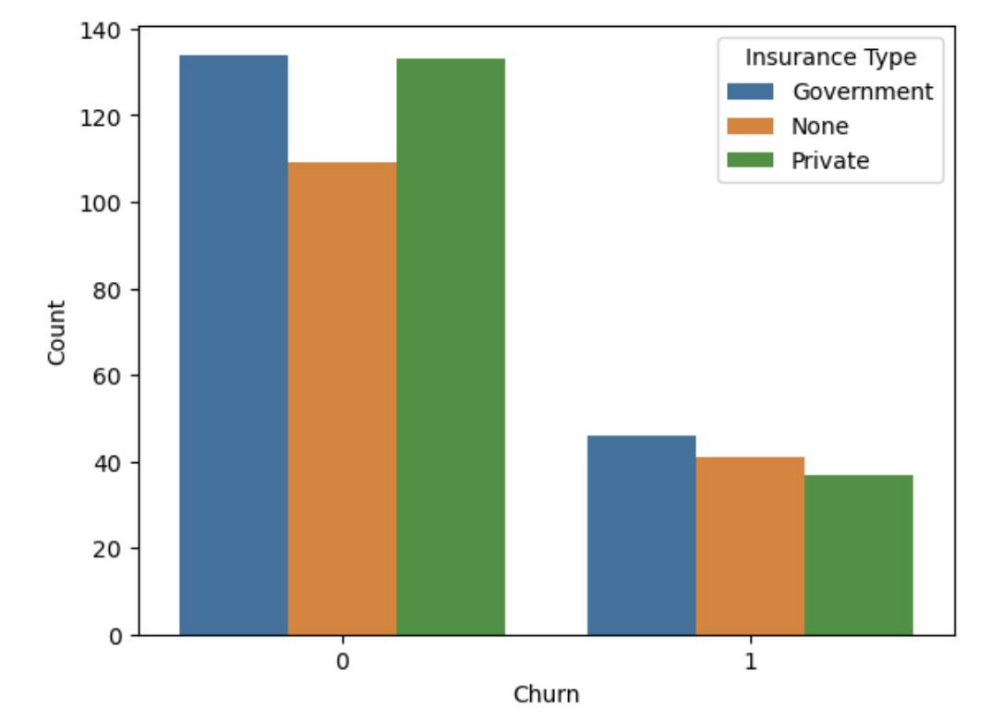
Exploratory Data Analysis

Potential factors influencing churn



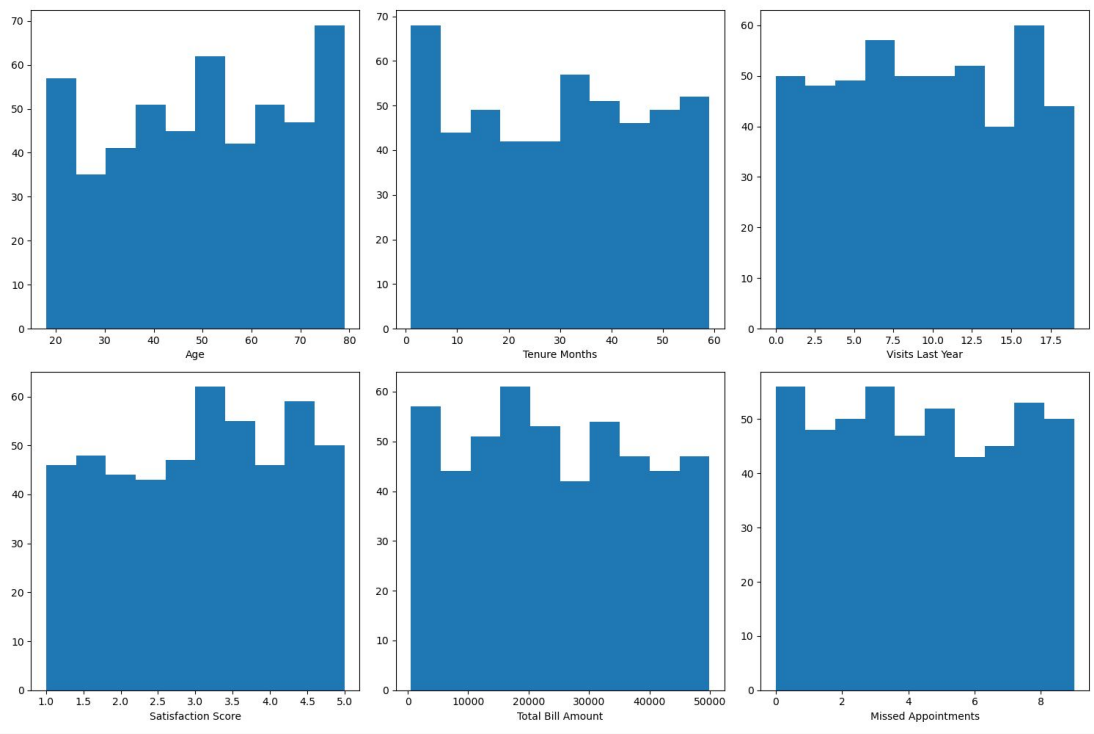
- Churned patients miss greater average number of visits than those that don't churn
- Churned patients have a greater average number of visits than those that retain
    - *Suggests more frequent appointments & health problems influences churn*

Potential factors influencing churn



- The percentage of churned patients with no insurance is higher than the uninsured percentage of their retained peer

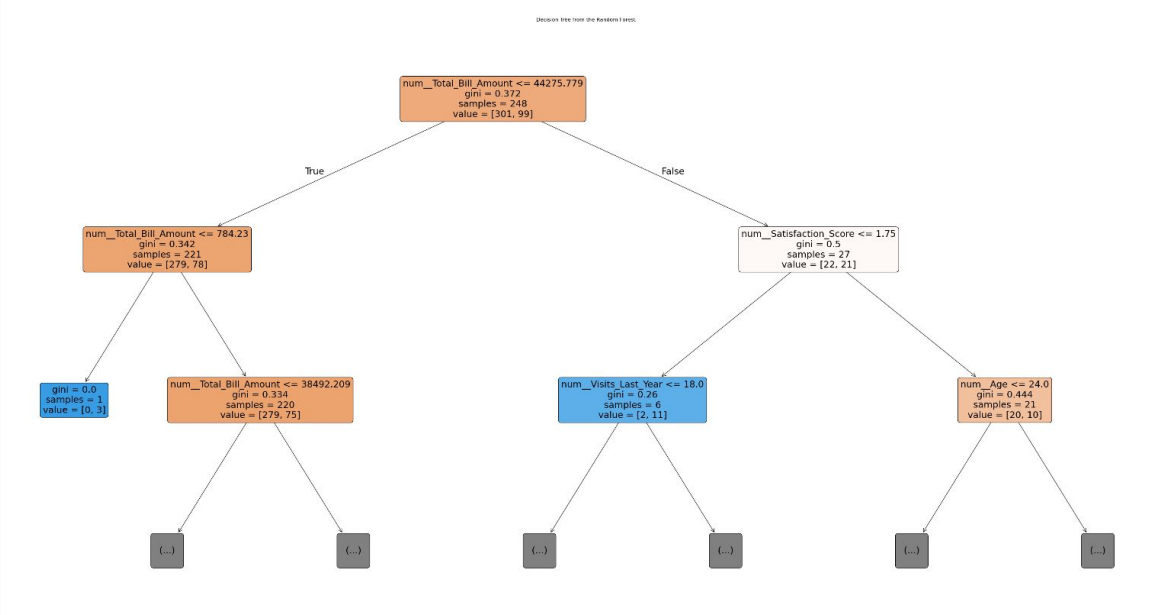
Distribution of Numeric Variables



- Variables somewhat uniformly distributed
- No apparent skew indicates limited need for logarithmic transformation

Results and Conclusions

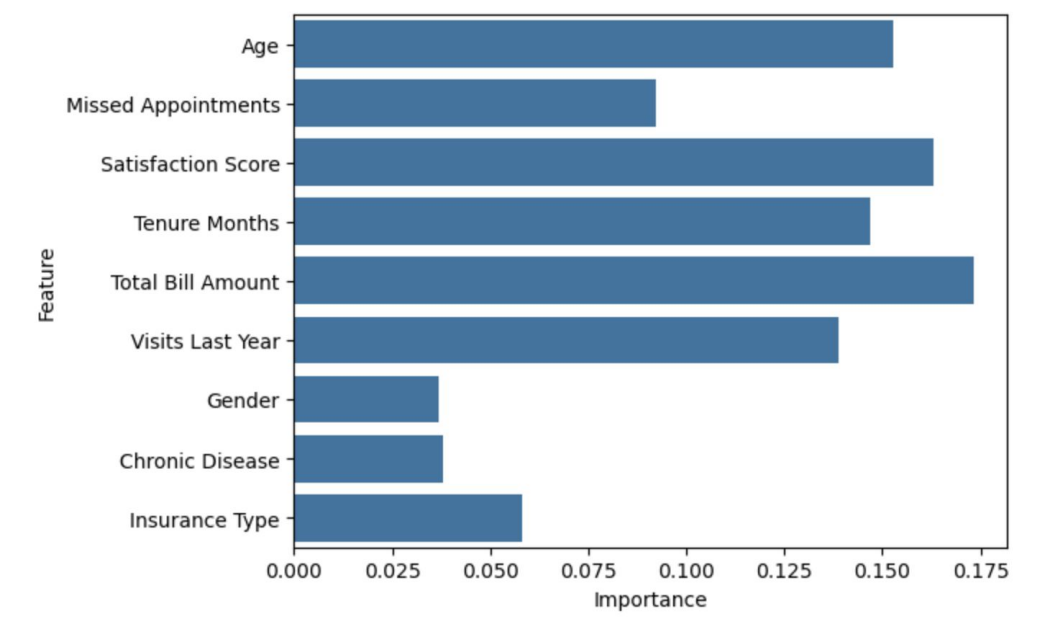
Random Forest Model



- Random Forest with 50 iterations and a max depth of 3
- CV grid search accuracy score: 0.755
- Test accuracy: 0.73

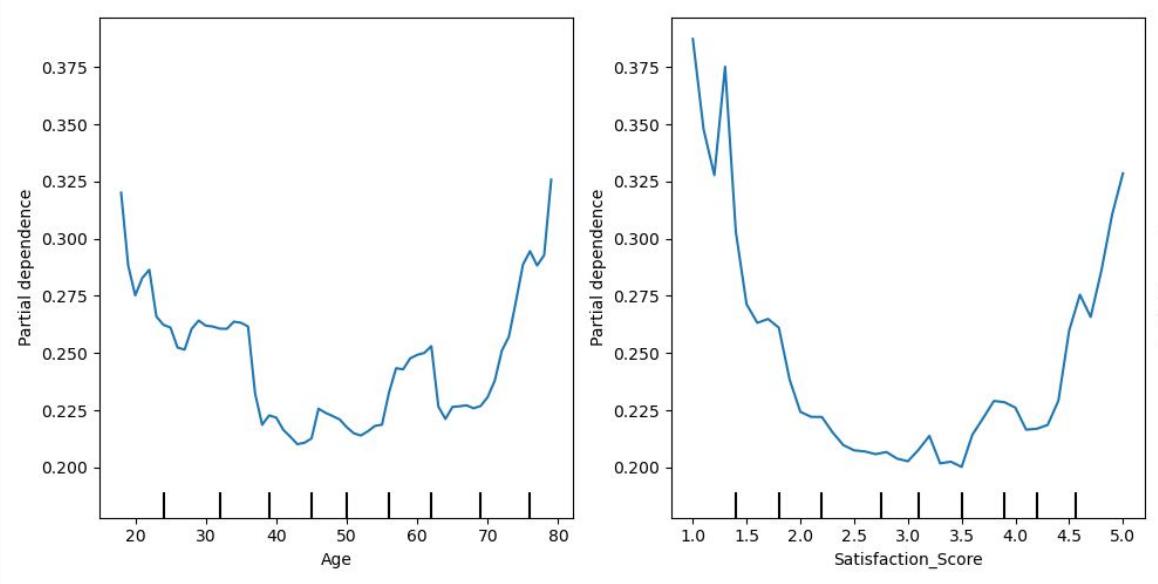
Key Summary: The RF model produced achieves 73% accuracy on new, unseen data

Feature Importances



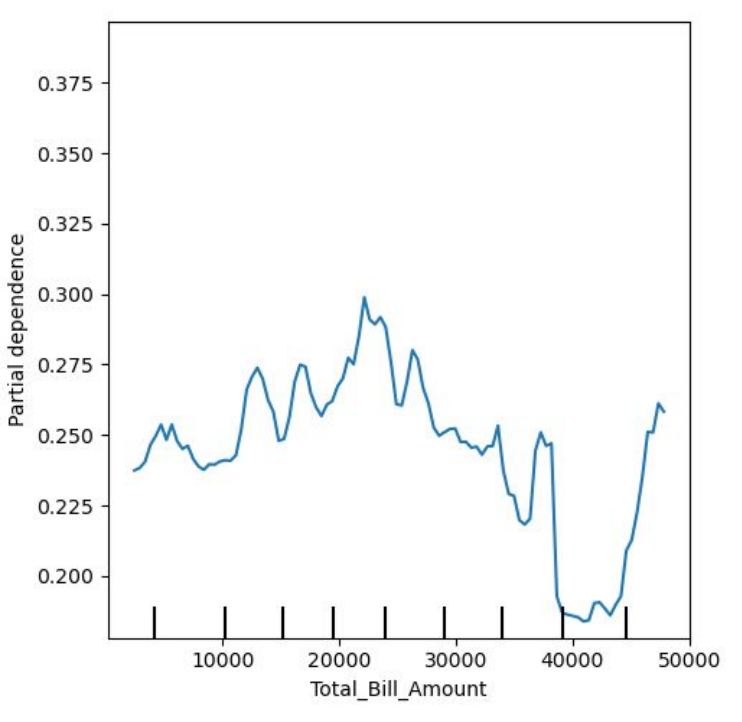
Age, Satisfaction Score, and Total Bill Amount are the most influential predictors of churn

Feature Importances



- As age increases, the predicted probability of churning *decreases* until ~70 years old, when the probability of churning sharply *increases*
  - *This may be because patients have passed on*
- As satisfaction score increases, the predicted probability of churn decreases, until satisfaction reaches 3.5/5.0, after which the probability of churn increases.
  - *This may be because patients are no longer suffering from the ailments that caused them to seek help initially*

Feature Importances



- As bill cost increases, the probability a patients will churn increases, for self-evident reasons
- After costs of $25,000, increased bill amount decreases the probability of churn
    - *This may reflect a correlation between the seriousness of an ailment, requiring multiple visits, and its cost*

Limitations

- With more time, I'd train a few different model types including xgboost and single vector machine models to evaluate the best model type for this task
- Partial Dependence Plots often mask interactions between variables; to rectify this would produce Individual Conditional Expectation to investigate the presence of feature interactions
- I'm intrigued by the PDP for Satisfaction Score which shows that the probability of churn increases after a score of 3.5/5.0. In future, I would investigate why greater satisfaction increases churn.